

```
In [5]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import norm
```

```
In [6]: df = pd.read_csv('E_Commerce.csv')
df
```

```
Out[6]:
```

	order_id	order_date	state	zone	category	brand_type	customer_
0	ORD0000001	2023-01-31	West Bengal	East	Fashion	Mass	
1	ORD0000002	2023-01-31	Gujarat	West	Sports & Fitness	Mass	
2	ORD0000003	2023-01-31	Delhi NCR	North	Grocery Essentials	Mass	
3	ORD0000004	2023-01-31	Madhya Pradesh	Central	Footwear	Mass	
4	ORD0000005	2023-01-31	Haryana	North	Fashion	Premium	
...	...	...	...	...	...	...	...
30595	ORD0030596	2025-12-31	Uttar Pradesh	North	Premium Lifestyle	Mass	
30596	ORD0030597	2025-12-31	Haryana	North	Footwear	Premium	
30597	ORD0030598	2025-12-31	Odisha	East	Footwear	Mass	
30598	ORD0030599	2025-12-31	West Bengal	East	Sports & Fitness	Mass	
30599	ORD0030600	2025-12-31	Maharashtra	West	Footwear	Premium	

30600 rows × 16 columns



1. An e-commerce company claims that the average discount percent offered is atleast 50%. Test whether the average discount percent is actually less that 50% at 5% level of significance

```
In [7]: x = df['discount_percent']
mu = 50
sigma = x.std(ddof=0) #ddof=0 for population deviation(sigma x) , ddof=1 for
n = len(x)
x_bar = x.mean()
z = (x_bar - mu) / (sigma/np.sqrt(n))
```

```
print('Z Calculated =' , z)
if z > -1.645:
    print('Accept H0')
else:
    print('Reject H0')
```

Z Calculated = -125.22041427648807
Reject H0

2. Test whether the mean final price differs between Male and Female customers at 5% level of significance

```
In [13]: male = df[df['customer_gender'] == 'Male']['final_price']
female = df[df['customer_gender'] == 'Female']['final_price']
x1 = male.mean()
x2 = female.mean()
s1 = male.std(ddof=0)
s2 = female.std(ddof=0)
n1 = len(male)
n2 = len(female)
z = (x1 - x2) / np.sqrt((s1**2/n1) + (s2**2/n2))
print('Calculate Z=', z)
if -1.96 < z < 1.96:
    print('Accept H0')
else:
    print('Reject H0')
```

Calculate Z= 0.9022301746506751
Accept H0

3. The company claims that less than 40% of customers purchase during Festival sales events. Test the claim at 5% level of significance

```
In [15]: success = df[df['sales_event'] == 'Festival']
n = len(df)
x = len(success)
p_hat = x / n
p0 = 0.40
z = (p_hat - p0) / np.sqrt((p0*(1-p0))/n)
print('Caluculated Z=', z)
if z < 1.645:
    print('Accept H0')
else:
    print('Reject H0')
```

Caluculated Z= -52.84890417135715
Accept H0

4. Test whether the proportion of Premium brand purchases differs between male and female customers at 5% level of significance

```
In [18]: male = df[df['customer_gender'] == 'Male']
female = df[df['customer_gender'] == 'Female']
x1 = len(male[male['brand_type'] == 'Premium'])
x2 = len(female[female['brand_type'] == 'Premium'])
n1 = len(male)
n2 = len(female)
p1 = x1 / n1
p2 = x2 / n2
p = (x1 + x2) / (n1 + n2)
q = 1 - p
z = (p1 - p2) / np.sqrt(p*q*((1/n1) + (1/n2)))
print('Calculated Z=', z)
if -1.96 < z < 1.96:
    print('Accept H0')
else:
    print('Reject H0')
```

Calculated Z= 1.876874162122772

Accept H0

In []: